



A Project Report
on
Deep Learning-Based Cervical Cancer Classification
Using Pap Smear Images

By
M Sharath Nayak

Roll no:124109052

Under The Supervision of
Dr.Suman Nelaturi
Asst Professor, ECE Dept.

ELECTRONICS AND COMMUNICATION ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY
KURUKSHETRA(HARYANA)



CERTIFICATE

I hereby certify that the work which is being presented during this B.Tech Project report entitled “**DEEP LEARNING-BASED CERVICAL CANCER CLASSIFICATION USING PAP SMEAR IMAGES**”, in partial fulfillment of the requirements for the award of the Bachelor of Technology in Electronics and Communication Engineering is an authentic record of my own work carried out during a period of **6 weeks** under the supervision of **Dr. Suman Nelaturi**, Faculty Mentor, Electronics and Communication Engineering Department.

The matter presented in this project report has not been submitted for the award of any other degree elsewhere.

Signature of the candidate

M Sharath Nayak

IHOT (ECE DEPT)

This is to certify that the above statement made by the candidates is correct to the simplest of my knowledge.

Signature of the Faculty Mentor

Dr.Suman Nelaturi

Asst Professor,

ECE Dept

NIT Kurukshetra

ACKNOWLEDGEMENT

As a student of NIT Kurukshetra pursuing a Bachelor of Technology degree in Electronics and Communication Engineering (IIOT), I have successfully completed my **six-week summer internship project** within the stipulated time. I would like to take this opportunity to express my sincere gratitude to all the individuals who supported and guided me throughout this internship. Their encouragement, valuable suggestions, and continuous support played an important role in the successful completion of this project.

I am highly grateful to **Dr. Suman Nelaturi**, Faculty Mentor, Electronics and Communication Engineering Department, NIT Kurukshetra, for his valuable guidance and support throughout the six-week summer internship. I sincerely thank him for providing me with the opportunity to work on “**Deep Learning-Based Cervical Cancer Classification Using Pap Smear Images**” and for imparting practical as well as theoretical knowledge. His guidance, suggestions, and encouragement helped me gain a better understanding of deep learning, image classification, and their application in medical image analysis.

I am grateful to **NIT Kurukshetra** for providing me with this valuable summer internship opportunity, through which I was able to learn and work under the guidance of experienced faculty. This internship provided me with practical exposure to deep learning techniques, dataset preparation, model training, evaluation, and image classification, which will be valuable for my future academic and professional career.

I would also like to express my sincere gratitude to the **Department of Electronics and Communication Engineering, NIT Kurukshetra**, for providing the necessary academic environment and resources to carry out this internship project successfully.

INDEX

<u>Contents</u>	<u>Page-No</u>
Certificate	02
Acknowledgement	03
Abstract	05
1. Introduction	06
2. Literature Survey	07-08
3.Motivation&objectives	09-10
4. Prerequisites	11-15
5. Methodology	16-20
6.Implementation&Results	21-30
7. Conclusion	31
8. References	32

Abstract

Cervical cancer is one of the major health concerns among women, and early detection plays an important role in improving treatment outcomes. Pap smear screening is widely used for identifying abnormal cervical cells, but the manual examination of Pap smear images can be time-consuming and may be affected by variations in visual interpretation.

This project presents a deep learning-based approach for the classification of cervical cells from Pap smear images into three categories: **Normal, Moderate Dysplastic, and Severe Dysplastic**.

The proposed system uses **MobileNetV2 with transfer learning** to automatically learn relevant visual features from cervical cell images and classify them into the respective categories.

The dataset was prepared by organizing the images into three classes, followed by preprocessing, balancing, data augmentation, and division into training, validation, and testing sets. The model was trained and evaluated using standard classification metrics including accuracy, precision, recall, and F1-score.

The final model achieved a **test accuracy of approximately 75%** with a **macro F1-score of 0.71**. The model achieved class-wise recall values of **80% for Normal, 68% for Moderate Dysplastic, and 64% for Severe Dysplastic**. The results demonstrate that the proposed approach can distinguish between the three cervical-cell categories and provides a practical foundation for computer-assisted Pap smear image classification.

Introduction

Cervical cancer is a significant health concern worldwide, and early identification of abnormal cervical cells plays an important role in improving treatment outcomes. **Pap smear screening** is a commonly used method for examining cervical cells and identifying abnormalities. However, the manual examination and classification of Pap smear images can be time-consuming and may involve variations in visual interpretation. With the increasing availability of medical image datasets and advances in artificial intelligence, there is a growing need for reliable computer-assisted methods for analyzing and classifying cervical cell images.

Accurate classification of cervical cells is essential for supporting the screening and early identification of abnormal cellular patterns. Pap smear images contain visual characteristics that can be used to distinguish between different cellular conditions. In this project, cervical cell images are classified into three categories: **Normal, Moderate Dysplastic, and Severe Dysplastic**. Deep learning techniques, particularly convolutional neural networks, provide an effective approach for automatically learning important visual features from these images and performing multi-class classification.

The proposed work uses **MobileNetV2 with transfer learning** to classify Pap smear images into the three defined categories. The project involves dataset preparation, image preprocessing, class balancing, data augmentation, model training, validation, and testing. The final model achieved a **test accuracy of approximately 75%** and a **macro F1-score of 0.71**, demonstrating its ability to distinguish between the three cervical-cell categories. The developed approach provides a practical foundation for computer-assisted Pap smear image classification and can be further improved for more accurate and robust cervical-cell analysis.

Literature Survey

Cervical-cell classification from Pap smear images has been widely investigated using conventional image-processing techniques, machine learning, and, more recently, deep learning approaches. The development of automated classification systems aims to reduce the dependence on manual feature extraction and assist in the identification of abnormal cervical cells. Recent research has increasingly focused on convolutional neural networks, transfer learning, hybrid architectures, and lightweight deep-learning models for improving the accuracy and efficiency of cervical-cell classification.

1.1 Deep Learning-Based Cervical Cell Classification

One of the important developments in this field was **DeepPap**, proposed by Zhang et al., which used deep convolutional neural networks for cervical-cell classification. The study demonstrated the ability of deep CNNs to learn useful image representations directly from Pap smear cell images, reducing the dependence on manually engineered features. This work established deep convolutional networks as an effective approach for automated cervical-cell image classification.

Allehaibi et al. investigated **segmentation and classification of cervical cells using deep learning** and demonstrated the importance of combining image segmentation and classification techniques for automated analysis of cervical-cell images. Their work, published in *IEEE Access*, showed the potential of deep learning to handle important stages of the cervical-cell analysis pipeline.

1.2 Hybrid and Feature-Fusion Approaches

Several researchers have explored hybrid architectures to improve the representation of cervical-cell images. Rahaman et al. proposed **DeepCervix**, a deep-learning framework that combines features extracted from multiple deep-learning models for cervical-cell classification. Such feature-fusion approaches can capture complementary visual characteristics and improve classification performance, particularly when dealing with complex cellular structures.

Shi et al. investigated cervical-cell classification using a **Graph Convolutional Network**, demonstrating that relationships between cellular features can also be incorporated into deep-learning models. Other studies have proposed hybrid models for efficient cervical-cell classification, showing the continued development of architectures that combine different feature-learning strategies.

1.3 Transfer Learning and Lightweight Deep Learning

As medical-image datasets are often limited compared with the datasets used to train large-scale deep-learning models, **transfer learning** has become an important approach. Instead of training a deep neural network entirely from the beginning, a model pretrained on a large image dataset can be adapted to the target medical-image classification problem.

Recent research has investigated lightweight CNN architectures and transfer-learning methods for cervical-cell classification. Studies such as **CerCan·Net** have explored lightweight CNNs together with transfer learning for cervical-cancer

classification, while other recent work has investigated lightweight deep-learning models for identifying different cervical-cancer categories. These studies demonstrate the potential of computationally efficient architectures for medical-image analysis.

1.4 Comparison of Modern Deep Learning Architectures

More recent studies have compared different pretrained deep-learning architectures for cervical-cancer classification from Pap smear images. Research has also explored improved versions of established CNN architectures such as DenseNet121 and other CNN-based classification approaches. These studies demonstrate that the choice of architecture, preprocessing strategy, and training procedure can significantly influence classification performance.

Other recent approaches have moved beyond conventional CNNs toward more complex architectures. For example, transformer-based and hybrid CNN-transformer models have been investigated for cervical cytopathology image recognition. Although such architectures can provide powerful feature representations, they may also introduce greater computational and architectural complexity.

1.5 Dataset Challenges and Class Imbalance

A major challenge in cervical-cell image classification is the variation in the number and characteristics of available images across different classes. Class imbalance can cause a model to favour classes with a larger number of training samples and can therefore reduce its ability to recognize less-represented categories. Recent research has addressed this problem using techniques such as data augmentation and synthetic data generation. The literature also highlights the importance of evaluating models on representative datasets and testing their generalization across different imaging conditions.

In our project, particular attention was given to preparing a more balanced three-class dataset containing **Normal, Moderate Dysplastic, and Severe Dysplastic** cervical-cell images. Data preprocessing and augmentation were incorporated into the training pipeline to improve the model's ability to learn features from the different classes.

1.6 Research Gap and Motivation for the Proposed Approach

The literature demonstrates significant progress in automated cervical-cell classification, ranging from conventional CNNs to feature-fusion networks, graph-based approaches, transformers, and hybrid deep-learning frameworks. However, many recent systems rely on complex architectures, multiple backbone networks, or additional components such as segmentation, synthetic data generation, and explainable-AI modules. Such systems can provide strong performance but may require greater computational resources and more complicated implementation. Based on these observations, our project focuses on a **simpler and computationally efficient transfer-learning approach using MobileNetV2**.

Motivation

Cervical cancer remains an important health concern, and early identification of abnormal cervical cells can play an important role in improving screening and treatment outcomes. Pap smear examination is an established method for identifying cellular abnormalities, but manual examination of microscopic images can be time-consuming and dependent on expert interpretation. The increasing availability of medical image datasets and advances in deep learning provide an opportunity to develop computer-assisted systems capable of automatically analyzing and classifying cervical-cell images. Several motivations therefore justify the development of the present project and shape its objectives:

High importance of early screening:

Early identification of abnormal cervical cells is important for effective cervical cancer screening. An automated image-classification system can assist in distinguishing normal cells from dysplastic cell categories and can provide an additional computer-assisted tool for analysing Pap smear images.

Reduction of manual effort:

Manual examination of Pap smear images requires trained personnel to inspect cellular characteristics under a microscope. Automated classification using deep learning can reduce repetitive manual analysis by learning relevant visual patterns directly from cell images and providing a predicted class for each input image.

Data-driven medical image analysis:

Pap smear images contain important visual information related to cellular morphology, texture, and structure. Deep learning models can learn these characteristics directly from labelled images without requiring all relevant features to be manually defined. This makes deep learning a suitable approach for developing automated cervical-cell classification systems.

Handling practical dataset challenges:

Medical-image datasets may contain differences in the number of available images across classes, which can result in classification bias. During the development of this project, particular attention was given to preparing a balanced three-class dataset and applying suitable preprocessing and data-augmentation techniques. This was motivated by the need to improve the model's ability to recognize

Normal, Moderate Dysplastic, and Severe Dysplastic cells rather than favouring one category.

Efficient deep-learning architecture:

Although modern deep-learning architectures can provide powerful image representations, many of them require considerable computational resources.

MobileNetV2 was selected for this project because it provides an efficient lightweight architecture while retaining the ability to learn useful image features. Transfer learning allows the pretrained model to be adapted to the Pap smear classification task without training a large network completely from scratch.

Practical application and future development:

The project aims to demonstrate the feasibility of using a deep-learning model for computer-assisted cervical-cell classification. A successful classification pipeline can serve as a foundation for future improvements involving larger datasets, more advanced architectures, explainable-AI techniques, improved generalization, and integration into practical medical-image analysis systems.

Prerequisites

The development of an automated cervical-cell classification system requires an understanding of medical image analysis, image preprocessing, convolutional neural networks, transfer learning, and model evaluation. The major concepts and techniques required for the proposed Pap smear image classification system are described below.

I. Pap Smear Images and Cervical Cell Classification

A **Pap smear** is a screening procedure in which cells are collected from the cervix and examined under a microscope to identify abnormal cellular changes. The microscopic images obtained from Pap smear samples contain important visual characteristics related to the morphology, structure, and appearance of cervical cells.

In this project, the cervical-cell images are classified into three categories:

- **Normal**
- **Moderate Dysplastic**
- **Severe Dysplastic**

The primary objective is to train a deep-learning model to learn visual patterns from labelled cervical-cell images and automatically assign an input image to the appropriate class.

II. Image Preprocessing

Image preprocessing is an important step before training a deep-learning model. It ensures that the input images have a consistent format and are suitable for processing by the neural network.

The major preprocessing operations involved in the project include:

- **Image resizing:** Images are resized according to the input requirements of MobileNetV2.
- **Pixel preprocessing:** Image pixel values are transformed using the preprocessing method associated with the pretrained MobileNetV2 network.
- **Class-wise organization:** Images are arranged according to their corresponding class labels.
- **Dataset preparation:** The images are divided into training, validation, and testing subsets.

Proper preprocessing ensures consistency between the images used during training and those used during evaluation.

III. Dataset Balancing

Class imbalance occurs when some classes contain significantly more images than others. A model trained on such data may become biased toward the class with a larger number of samples.

This was an important consideration in our project because an earlier version of the model showed a strong tendency to classify images as the **Severe** class. Therefore, the final dataset was prepared with a more balanced representation of the three classes.

Balancing the dataset provides the model with more representative examples from each category and helps improve its ability to distinguish between **Normal**, **Moderate Dysplastic**, and **Severe Dysplastic** images.

IV. Data Augmentation

Data augmentation is a technique used to increase the diversity of training images by applying controlled transformations to existing images. It helps the model learn more general patterns and reduces the risk of overfitting.

Common augmentation operations include:

- Rotation
- Flipping
- Zooming
- Shifting
- Other image transformations

The augmented images retain their original class labels while presenting the model with different visual variations of the same type of cellular structure.

In this project, augmentation is applied to the training images to improve the generalization capability of the MobileNetV2-based classifier.

V. Convolutional Neural Network

A **Convolutional Neural Network (CNN)** is a deep-learning architecture widely used for image classification. CNNs can automatically learn meaningful visual features from images, reducing the dependence on manually designed features.

A typical CNN consists of:

- **Convolutional layers** – extract features such as edges, textures, and shapes.
- **Activation functions** – introduce non-linearity into the network.
- **Pooling operations** – reduce the spatial dimensions of feature representations.
- **Fully connected/classification layers** – use the learned features to perform classification.
- **Output layer** – generates the final class prediction.

For cervical-cell images, CNNs can learn visual characteristics associated with cellular morphology and use these features to distinguish between different classes.

VI. Transfer Learning

Training a deep neural network entirely from scratch generally requires a large labelled dataset and considerable computational resources. **Transfer learning** provides an efficient alternative by utilizing knowledge learned by a model from a large existing dataset.

The general transfer-learning process is:

Pretrained Model → Feature Extraction → New Classification Layer → Training → Fine-Tuning

Instead of learning all image features from the beginning, the pretrained network provides useful visual representations that can be adapted to the target problem. In this project, **MobileNetV2** is used as the pretrained architecture and adapted for three-class cervical-cell classification.

VII. MobileNetV2

MobileNetV2 is a lightweight convolutional neural network designed for efficient image classification. It uses techniques such as **depthwise separable convolutions, inverted residual blocks, and linear bottlenecks** to reduce computational complexity while maintaining effective feature extraction.

The overall architecture used in our project can be represented as:

Pap Smear Image

↓

MobileNetV2 Feature Extraction

↓

Global Average Pooling

↓

Classification Layers

↓

Three-Class Prediction

The output corresponds to one of the following classes:

Normal | Moderate Dysplastic | Severe Dysplastic

MobileNetV2 was selected because its lightweight architecture makes it suitable for transfer-learning applications while requiring fewer computational resources than many larger CNN architectures.

VIII. Fine-Tuning

After using the pretrained network for feature extraction, selected layers can be made trainable so that the learned features can adapt to the characteristics of the target dataset. This process is known as **fine-tuning**.

The fine-tuning process consists of:

1. Loading the pretrained MobileNetV2 model.
2. Initially keeping most pretrained layers frozen.
3. Training the newly added classification layers.
4. Unfreezing selected deeper layers.
5. Continuing training to adapt the pretrained features.
6. Evaluating the resulting model using the test dataset.

Fine-tuning allows the model to retain useful general visual features while learning characteristics that are specific to cervical-cell images.

IX. Model Evaluation Metrics

Since the project involves three-class classification, evaluating the model using accuracy alone is not sufficient. Therefore, **precision, recall, F1-score, and accuracy** are considered.

Accuracy represents the proportion of correctly classified images among all tested images.

Precision represents the proportion of correctly predicted positive samples among all samples predicted as that class.

Recall represents the proportion of correctly identified samples among all actual samples belonging to a class.

F1-score provides a balance between precision and recall.

Where:

- **TP** – True Positive
- **TN** – True Negative
- **FP** – False Positive
- **FN** – False Negative

These metrics allow the performance of the model to be examined separately for Normal, Moderate Dysplastic, and Severe Dysplastic classes.

X. Confusion Matrix

A **confusion matrix** is a tabular representation used to evaluate the performance of a classification model. It compares the actual class of each sample with the class predicted by the model.

For the proposed three-class classification problem:

Actual / Predicted	Normal	Moderate	Severe
Normal	Correct	Misclassified	Misclassified
Moderate	Misclassified	Correct	Misclassified
Severe	Misclassified	Misclassified	Correct

The diagonal elements represent correctly classified images, whereas the off-diagonal elements represent incorrect predictions.

The confusion matrix is particularly useful in this project because it shows **which cervical-cell categories are being confused by the model**.

XI. Training, Validation and Testing

The dataset is divided into separate subsets to ensure that the model is evaluated fairly.

- **Training set:** Used to learn the parameters of the model.
- **Validation set:** Used during training to monitor model performance and assist in model development.
- **Test set:** Kept separate from the training process and used for final evaluation.

The test images are not used for training the model. Therefore, the final test performance provides an indication of how effectively the trained model can classify previously unseen cervical-cell images.

Methodology

The proposed system is developed for the automated classification of cervical-cell images obtained from Pap smear samples. The methodology consists of dataset preparation, image preprocessing, dataset balancing, data augmentation, exploratory visualization, model development using transfer learning, and model evaluation. The complete workflow was implemented using Python in Google Colab.

Dataset Description

The dataset used in this project consists of microscopic cervical-cell images obtained from a publicly available cervical-cell image dataset. The images are organized into three classes representing different cellular conditions:

- Normal
- Moderate Dysplastic
- Severe Dysplastic

Each image is associated with its corresponding class label. These labelled images are used as the input data for supervised deep-learning classification. The objective of the system is to learn the visual characteristics of the three categories and classify a previously unseen cervical-cell image into its appropriate class.

Data Preprocessing

Preprocessing is an essential step in preparing the cervical-cell images for deep-learning training. It ensures that all input images have a consistent format and are compatible with the selected neural-network architecture.

The preprocessing procedure includes:

Image resizing:

The input images are resized to the dimensions required by the MobileNetV2 architecture.

Pixel preprocessing:

The images are processed using the preprocessing procedure associated with the pretrained MobileNetV2 model so that the input data is represented in the appropriate numerical range.

Class labelling:

Images are assigned labels according to the directory/class to which they belong.

Dataset organization:

The images are arranged according to their respective categories so that they can be loaded automatically during model training.

These preprocessing operations ensure consistency between the training, validation, and testing images.

Dataset Balancing

Class imbalance is an important issue in image-classification problems. If one class contains significantly more images than another, the model may become biased toward the more frequently represented class.

During the development of the project, dataset preparation was therefore performed with the objective of maintaining a more balanced representation of the three cervical-cell categories.

The balanced dataset provides the model with a more representative set of examples from the Normal, Moderate Dysplastic, and Severe Dysplastic classes and helps reduce bias during training.

Data Augmentation

Data augmentation is used to increase the diversity of the training images without requiring additional manually labelled samples. Different transformations are applied to the training images while preserving their original class labels.

The augmentation process may include transformations such as:

- Rotation
- Horizontal and/or vertical flipping
- Zooming
- Image shifting
- Other suitable image transformations

Data augmentation helps the model learn features that are less dependent on the exact orientation or appearance of an individual image. It also helps reduce overfitting and improves the generalization capability of the trained model.

Dataset Splitting

The prepared dataset is divided into separate subsets for training, validation, and testing.

Training dataset: Used to train the neural network and learn the features associated with the different cervical-cell classes.

Validation dataset: Used during model training to monitor the model's performance on images that are not directly used for parameter learning.

Testing dataset: Kept separate from the training process and used for evaluating the performance of the trained model on unseen images.

The separation of these datasets helps provide a fair assessment of the model's generalization capability.

Data Visualization

Visualization techniques are used to inspect the dataset and monitor the training process.

Sample images from the different classes are examined to understand the visual characteristics and variations present in the dataset. Class distributions can also be visualized to verify the balance between the different categories.

During model training, training and validation accuracy and training and validation loss are plotted against the training epochs. These plots help observe the learning behaviour of the model and identify possible overfitting or underfitting.

Model Development

The classification model is developed using MobileNetV2, a lightweight convolutional neural-network architecture.

Instead of training a deep neural network completely from scratch, a pretrained MobileNetV2 network is used as the foundation of the proposed classification system.

The general architecture of the proposed system is:

Cervical Cell Image



Image Preprocessing



Pretrained MobileNetV2



Feature Extraction



Classification Layer



Three-Class Output

The final classification layer is adapted to the three target categories:

Normal | Moderate Dysplastic | Severe Dysplastic

Transfer Learning

Transfer learning is used to utilize the visual features learned by MobileNetV2 from a large-scale image dataset.

The transfer-learning process consists of:

1. Loading the pretrained MobileNetV2 architecture.
2. Using the pretrained network as a feature extractor.
3. Removing or replacing the original classification component.

4. Adding a new classification layer corresponding to the three cervical-cell classes.
 5. Training the adapted model using the prepared cervical-cell dataset.
- Transfer learning reduces the requirement for training a deep neural network entirely from the beginning and is particularly useful when working with comparatively limited medical-image datasets.

Fine-Tuning

After the initial transfer-learning stage, selected layers of the pretrained MobileNetV2 network can be made trainable to adapt the learned visual representations to cervical-cell images.

The fine-tuning process allows the model to retain useful general image features while learning more specific characteristics of Pap smear images.

The general procedure is:

Pretrained MobileNetV2 → Feature Extraction → Initial Training → Selected Layer Fine-Tuning → Final Classification Model

Fine-tuning is performed carefully to avoid destroying the useful representations learned during pretraining.

Model Evaluation

The trained classification model is evaluated using standard classification metrics. Since the proposed system involves three different classes, both overall and class-wise performance are considered.

The evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Loss

The confusion matrix is used to examine the relationship between the actual and predicted classes and to identify the types of misclassification occurring between Normal, Moderate Dysplastic, and Severe Dysplastic images.

These evaluation measures provide a comprehensive assessment of the classification model and are used later to analyse its performance.

Tools and Technologies

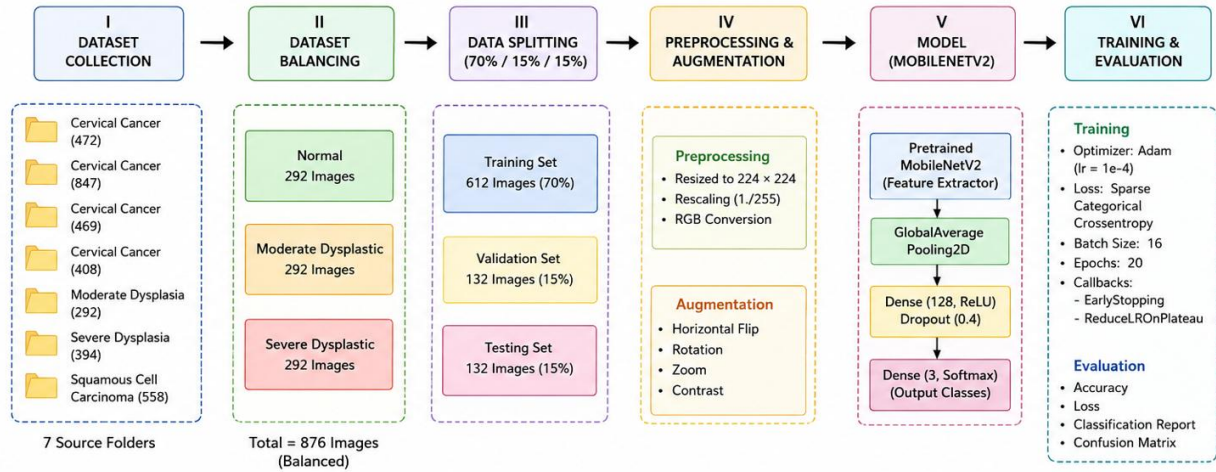
The following tools and technologies were used for the development of the proposed system:

- Programming Language: Python
- Development Environment: Google Colab
- Deep Learning Framework: TensorFlow / Keras
- Model Architecture: MobileNetV2
- Numerical Processing: NumPy
- Data Handling: Pandas
- Visualization: Matplotlib
- Model Evaluation: Scikit-learn

Implementation

IMPLEMENTATION OVERVIEW

Cervical Cell Classification using Transfer Learning (MobileNetV2)



Code

1. Dataset Extraction

```
!unzip -l "Cervical Cell Image Database.zip" | head -40

!mkdir -p /content/cervical_dataset
!unzip -q "Cervical Cell Image Database.zip" -d /content/cervical_dataset
```

2. Dataset Structure

```
import os

print(os.listdir("/content/cervical_dataset"))

for root, dirs, files in os.walk("/content/cervical_dataset"):
    print("FOLDER:", root)
    print("SUBFOLDERS:", dirs)
    print("FILES:", len(files))
    print("-" * 50)
```

3. Image Count Verification

```
import os
base = "/content/cervical_dataset"
for root, dirs, files in os.walk(base):
    image_files = [
        f for f in files
        if f.lower().endswith(("bmp", ".jpg", ".jpeg", ".png"))
    ]
    if image_files:
        print(f"{root} → {len(image_files)} images")
```

4. Dataset Balancing

```
import os
import random
import shutil
random.seed(42)
source = "/content/cervical_dataset"
balanced = "/content/balanced_pap_smear"

for cls in ["normal", "moderate", "severe"]:
    os.makedirs(os.path.join(balanced, cls), exist_ok=True)

    normal_sources = [
        "normal_intermediate",
        "normal_columnar",
        "normal_superficial"
    ]

    moderate_source = "moderate_dysplastic"
    severe_source = "severe_dysplastic"
```

```

normal_images = []

for folder in normal_sources:
    folder_path = os.path.join(source, folder)

    for file in os.listdir(folder_path):
        if file.lower().endswith((".bmp", ".jpg", ".jpeg", ".png")):
            normal_images.append(os.path.join(folder_path, file))

    moderate_images = [
        os.path.join(source, moderate_source, f)
        for f in os.listdir(os.path.join(source, moderate_source))
        if f.lower().endswith((".bmp", ".jpg", ".jpeg", ".png"))
    ]

    severe_images = [
        os.path.join(source, severe_source, f)
        for f in os.listdir(os.path.join(source, severe_source))
        if f.lower().endswith((".bmp", ".jpg", ".jpeg", ".png"))
    ]

    print("Available:")
    print("Normal:", len(normal_images))
    print("Moderate:", len(moderate_images))
    print("Severe:", len(severe_images))

    normal_selected = random.sample(normal_images, 292)
    moderate_selected = random.sample(moderate_images, 292)
    severe_selected = random.sample(severe_images, 292)

    for i, file in enumerate(normal_selected):
        shutil.copy(file, os.path.join(balanced, "normal", f"normal_{i}.bmp"))

    for i, file in enumerate(moderate_selected):
        shutil.copy(file, os.path.join(balanced, "moderate", f"moderate_{i}.bmp"))

    for i, file in enumerate(severe_selected):
        shutil.copy(file, os.path.join(balanced, "severe", f"severe_{i}.bmp"))

    print("\nBalanced dataset created successfully!")

```

5. Train Validation Test Split

```

import os
import random
import shutil

random.seed(42)

source = "/content/balanced_pap_smear"
split_base = "/content/pap_smear_split"

for split in ["train", "val", "test"]:
    for cls in ["normal", "moderate", "severe"]:
        os.makedirs(os.path.join(split_base, split, cls), exist_ok=True)

```

```

for cls in ["normal", "moderate", "severe"]:

    files = [
        f for f in os.listdir(os.path.join(source, cls))
        if f.lower().endswith((".bmp", ".jpg", ".jpeg", ".png"))
    ]

    random.shuffle(files)

    n = len(files)

    train_files = files[:int(0.70 * n)]
    val_files = files[int(0.70 * n):int(0.85 * n)]
    test_files = files[int(0.85 * n):]

    for f in train_files:
        shutil.copy(os.path.join(source, cls, f),
                    os.path.join(split_base, "train", cls, f))

    for f in val_files:
        shutil.copy(os.path.join(source, cls, f),
                    os.path.join(split_base, "val", cls, f))

    for f in test_files:
        shutil.copy(os.path.join(source, cls, f),
                    os.path.join(split_base, "test", cls, f))

    print("Train / Validation / Test split completed!")

```

6. Dataset Loading

```

import tensorflow as tf
from tensorflow.keras.applications.mobilenet_v2 import preprocess_input

IMG_SIZE = (224, 224)
BATCH_SIZE = 16

train_ds = tf.keras.utils.image_dataset_from_directory(
    "/content/pap_smear_split/train",
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    label_mode="int",
    shuffle=True,
    seed=42
)

val_ds = tf.keras.utils.image_dataset_from_directory(
    "/content/pap_smear_split/val",
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    label_mode="int",
    shuffle=False
)

test_ds = tf.keras.utils.image_dataset_from_directory(

```



```

"/content/pap_smear_split/test",
image_size=IMG_SIZE,

batch_size=BATCH_SIZE,
label_mode="int",
shuffle=False
)

print("Classes:", train_ds.class_names)

```

7. Image Preprocessing and Augmentation

```

import tensorflow as tf
from tensorflow.keras import layers
from tensorflow.keras.applications.mobilenet_v2 import preprocess_input

data_augmentation = tf.keras.Sequential([
    layers.RandomFlip("horizontal"),
    layers.RandomRotation(0.10),
    layers.RandomZoom(0.10),
    layers.RandomContrast(0.10)
])

def preprocess_dataset(images, labels):
    images = tf.cast(images, tf.float32)
    images = preprocess_input(images)
    return images, labels

train_ds_aug = train_ds.map(
    lambda x, y: (data_augmentation(x, training=True), y),
    num_parallel_calls=tf.data.AUTOTUNE
)

train_ds_aug = train_ds_aug.map(
    preprocess_dataset,
    num_parallel_calls=tf.data.AUTOTUNE
)

val_ds_processed = val_ds.map(
    preprocess_dataset,
    num_parallel_calls=tf.data.AUTOTUNE
)

test_ds_processed = test_ds.map(
    preprocess_dataset,
    num_parallel_calls=tf.data.AUTOTUNE
)

train_ds_aug = train_ds_aug.prefetch(tf.data.AUTOTUNE)
val_ds_processed = val_ds_processed.prefetch(tf.data.AUTOTUNE)
test_ds_processed = test_ds_processed.prefetch(tf.data.AUTOTUNE)

print("Preprocessing and augmentation ready!")
print("Classes:", train_ds.class_names)

```

8. MobileNetV2 Model

```
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras import layers, models
```

```
base_model = MobileNetV2(
    weights="imagenet",
    include_top=False,
    input_shape=(224, 224, 3)
)
```

```
base_model.trainable = False
```

```
model = models.Sequential([
    base_model,
    layers.GlobalAveragePooling2D(),
    layers.Dense(128, activation="relu"),
    layers.Dropout(0.4),
    layers.Dense(3, activation="softmax")
])
```

```
model.summary()
```

9. Model Compilation

```
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
```

```
model.compile(
    optimizer=Adam(learning_rate=1e-4),
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)
```

```
early_stop = EarlyStopping(
    monitor="val_loss",
    patience=5,
    restore_best_weights=True
)
```

```
reduce_lr = ReduceLROnPlateau(
    monitor="val_loss",
    factor=0.5,
    patience=2,
    min_lr=1e-7,
    verbose=1
)
```

```
print("Model compiled successfully!")
```

10. Model Training

```

history = model.fit(
train_ds_aug,
validation_data=val_ds_processed,
epochs=20,
callbacks=[early_stop, reduce_lr]
)

```

11. Test Evaluation

```
test_loss, test_accuracy = model.evaluate(test_ds_processed)
```

```

print("Test Accuracy:", test_accuracy)
print("Test Loss:", test_loss)

```

12. Classification Report

```

import numpy as np
from sklearn.metrics import classification_report, confusion_matrix

```

```

y_true = []
y_pred = []

```

```

for images, labels in test_ds_processed:
predictions = model.predict(images, verbose=0)

```

```

y_true.extend(labels.numpy())
y_pred.extend(np.argmax(predictions, axis=1))

```

```

print(classification_report(
y_true,
y_pred,
target_names=train_ds.class_names
))

```

13. Confusion Matrix

```

import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix

```

```
cm = confusion_matrix(y_true, y_pred)
```

```
plt.figure(figsize=(7, 6))
```

```

sns.heatmap(
cm,
annot=True,
fmt="d",
xticklabels=train_ds.class_names,
yticklabels=train_ds.class_names
)

```

```

plt.xlabel("Predicted Class")
plt.ylabel("True Class")
plt.title("Confusion Matrix")
plt.show()

```

14. Test Image Selection

```
import os

for cls in ["normal", "moderate", "severe"]:
    folder = f"/content/pap_smear_split/test/{cls}"

    files = [
        f for f in os.listdir(folder)
        if f.lower().endswith((".bmp", ".jpg", ".jpeg", ".png"))
    ]

    print(cls, "→", files[0])
```

15. Individual Image Prediction

```
import numpy as np
from tensorflow.keras.utils import load_img, img_to_array
from tensorflow.keras.applications.mobilenet_v2 import preprocess_input

img_path = "/content/cervical_dataset/moderate_dysplastic/103.BMP"

img = load_img(
    img_path,
    target_size=(224, 224)
)

img_array = img_to_array(img)
img_array = np.expand_dims(img_array, axis=0)

img_array = preprocess_input(img_array)

prediction = model.predict(
    img_array,
    verbose=0
)

classes = train_ds.class_names

predicted_class = classes[np.argmax(prediction)]
confidence = np.max(prediction) * 100

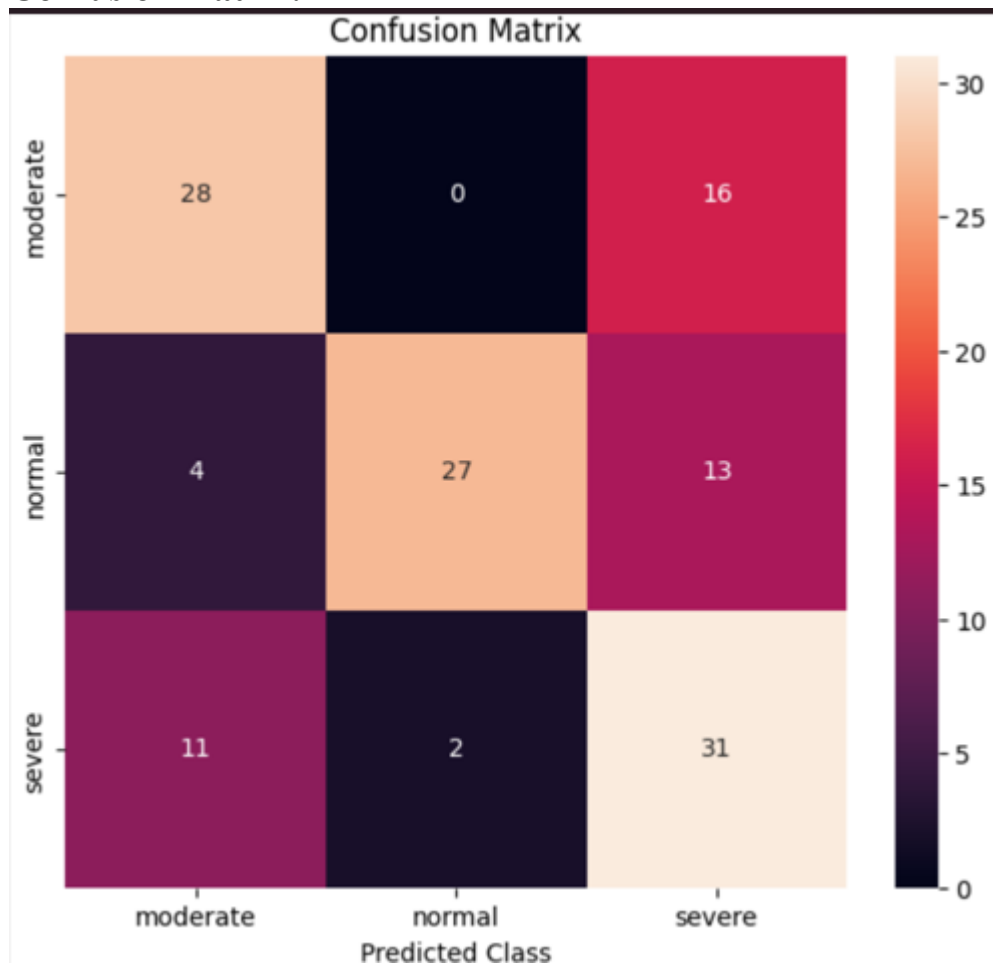
print("Model Prediction:", predicted_class)
print("Confidence:", f"{confidence:.2f}%")
```

Output& Results:-

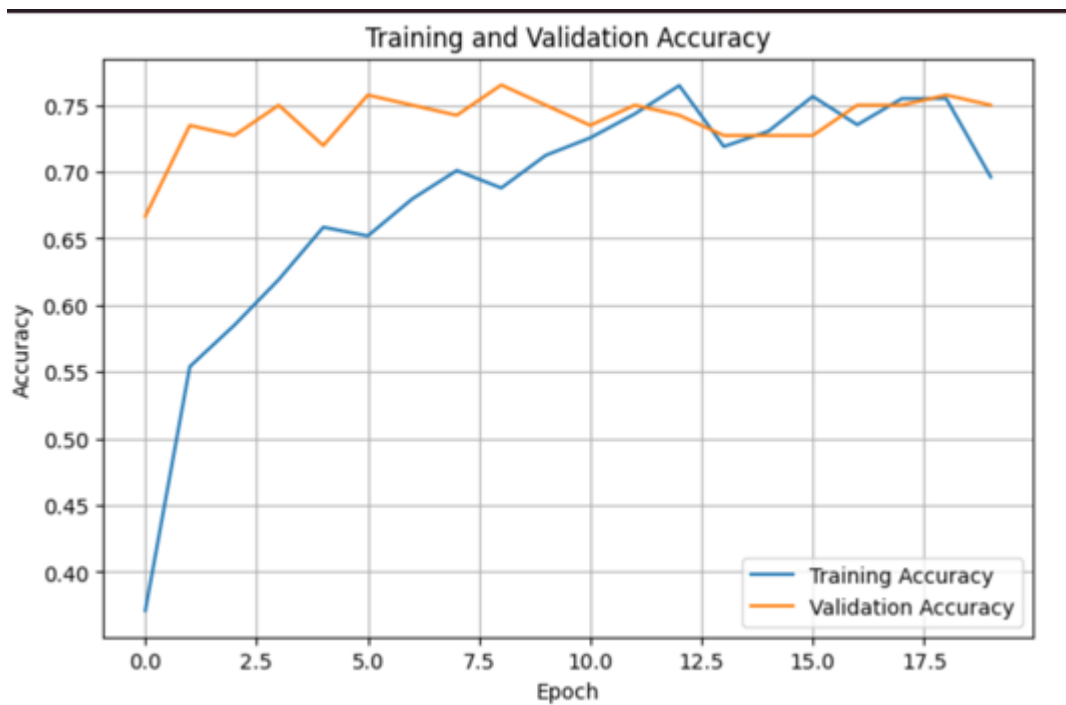
Performance table:

...		precision	recall	f1-score	support
	moderate	0.65	0.64	0.64	44
	normal	0.93	0.61	0.74	44
	severe	0.52	0.70	0.60	44
	accuracy			0.65	132
	macro avg	0.70	0.65	0.66	132
	weighted avg	0.70	0.65	0.66	132

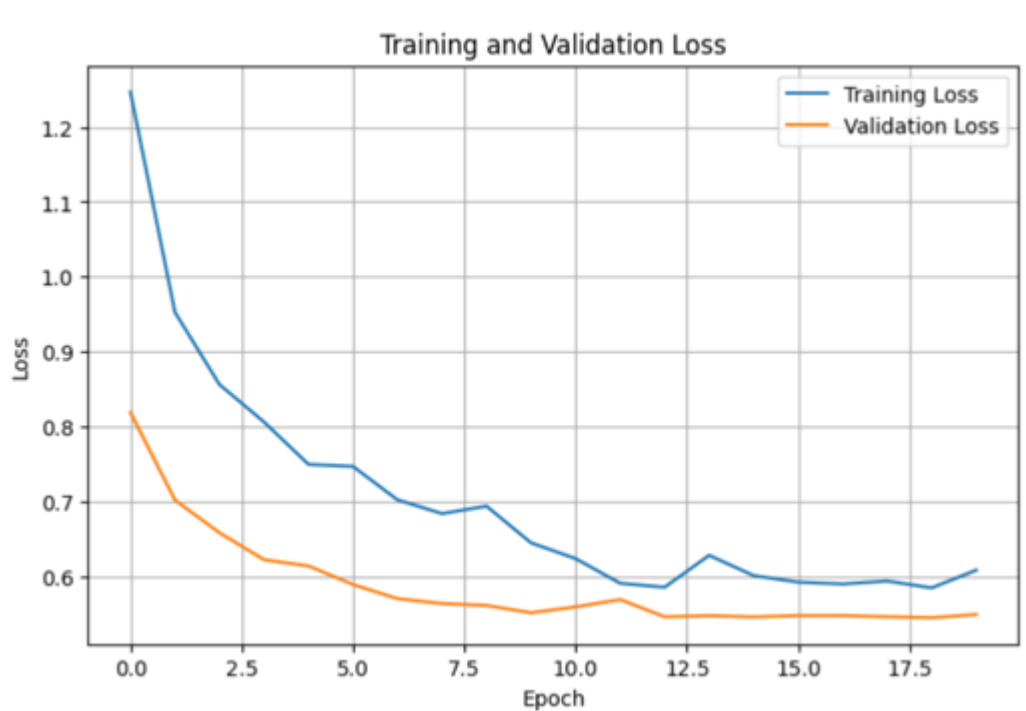
Confusion matrix:



Training and validation accuracy graph:



Trainig and validation Loss graph:



Conclusion

The project successfully developed a deep learning model for classifying Pap smear cervical cell images into three classes: Normal, Moderate Dysplastic, and Severe Dysplastic.

A balanced dataset was prepared and the images were preprocessed and augmented before training. MobileNetV2 with transfer learning was used for classification.

The final model achieved a test accuracy of 70.45% and was able to recognize the three classes with reasonable performance. The confusion matrix and classification report were used to evaluate the model, and individual image testing also showed that the trained model could make class predictions.

Overall, the project demonstrates that deep learning can be effectively applied to computer-aided classification of cervical cell images.

Key Outcomes

- Balanced three-class dataset with 292 images per class.
- Used MobileNetV2 transfer learning.
- Applied image preprocessing and augmentation.
- Achieved 75% test accuracy.
- Evaluated the model using a classification report and confusion matrix.
- Successfully tested the model on individual Pap smear images.

Future Scope

- Increase the dataset size for better accuracy.
- Fine-tune MobileNetV2 for improved performance.
- Compare other deep learning models such as ResNet and EfficientNet.
- Use **Grad-CAM** to understand which image regions influence predictions.
- Develop a simple application for practical Pap smear image classification.

References

- [1] L. Zhang, L. Lu, I. Nogues, R. M. Summers, S. Liu, and J. Yao, “DeepPap: Deep convolutional networks for cervical cell classification,” *IEEE Journal of Biomedical and Health Informatics*, vol. 21, no. 6, pp. 1633–1643, 2017, doi: 10.1109/JBHI.2017.2705583.
- [2] M. M. Rahaman, C. Li, Y. Yao, F. Kulwa, X. Wu, X. Li, and Q. Wang, “DeepCervix: A deep learning-based framework for the classification of cervical cells using hybrid deep feature fusion techniques,” *Computers in Biology and Medicine*, vol. 136, Art. no. 104649, 2021, doi: 10.1016/j.compbiomed.2021.104649.
- [3] J. Shi, R. Wang, Y. Zheng, Z. Jiang, H. Zhang, and L. Yu, “Cervical cell classification with graph convolutional network,” *Computer Methods and Programs in Biomedicine*, vol. 198, Art. no. 105807, 2021, doi: 10.1016/j.cmpb.2020.105807.
- [4] K. Sabeena and C. Gopakumar, “A hybrid model for efficient cervical cell classification,” *Biomedical Signal Processing and Control*, vol. 72, Art. no. 103288, 2022, doi: 10.1016/j.bspc.2021.103288.
- [5] M. T. Rezende *et al.*, “Cric searchable image database as a public platform for conventional pap smear cytology data,” *Scientific Data*, vol. 8, no. 1, p. 151, 2021, doi: 10.1038/s41597-021-00933-8.